Homework 1
Image Classification

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Introduction

- For this assignment, you are required to build a neural network with **PyTorch** and train it to carry out a **classification task.**
- To ensure that students can meet the assignment's requirements, **the use of pretrained weights and existing models is not permitted.**

What to do?

- You're tasked with completing at least 3 Python Files: ***model.py, train.py, test.py***
- You need to implement your neural network inside ***model.py*** and named it as "ClassificationModel". This network should be rendered accessible to ***train.py*** and ***test.py*** through import.
- Within these files, you can import any package and design any additional classes or functions if you need. However, the utilization of ready-made neural network and pre-trained weight is forbidden.
- **Discovery of any infringement of this cardinal rule will incur a penalty of a zero score for this assignment.**

Dataset

1. The dataset consists of sports images from **100 different categories.** The size of each image is **224*224*3.**
2. A total of 10000+ images are provided for training.
3. **No validation set is given.** Please split the training set yourself if you wish to evaluate your model's performance.

Grading (Performance 90% + Report 10%)

- ***Top-5 Accuracy (75 points)***

  65% <= Accuracy, get full points

  60% <= Accuracy < 65%, get 65 points

  55% <= Accuracy < 60%, get 55 points

  50% <= Accuracy < 55%, get 45 points

  Accuracy < 50%, get 0 points

- ***Number of Parameters (15 points)***

  This evaluation is based on the number of parameter of your model. The fewer parameters you use, the higher score you will get.

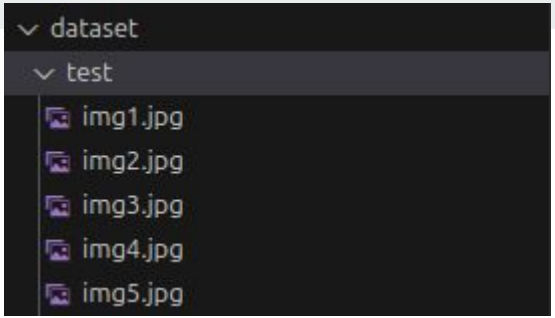
  Score formula: $\text{Round}\left(\frac{n-r}{n-1} \times 15\right)$

  n: number of students

  r: your ranking (r = 1 being the best rank)

Report

- In the report, briefly describe your approach.
- e.g. the design of the model architecture, strategies for balancing model performance with computational and parameter efficiency, and the development of an effective training strategy



Grading

- TA will use ***test.py*** on a test set to generate a CSV file for final grading. Make sure your model works with the script. Final grading will use the CSV generated from a hidden test set.
- For grading, the TA will only modify the DATAPATH variable in ***test.py*** to point to the hidden test dataset—**do not hard-code this path**; your code must read the data path from DATAPATH.

```
DATAPATH = "./dataset/test"
# We would ONLY change this path to test your model
# Please make sure you use this variable to specify the data path in your code
```

- You can use the given file ***weight.py*** to evaluate your model parameters.

```
(base) syc@sycRTX:~/Desktop/VST_113/TA$ python weight.py
# parameters: 15556
```

Grading

- After *cd* to folder, and change the data path, **TA will execute "python test.py“.**
- Your ***test.py*** need to generate **"pred_{student_id}.csv"** based on the weight(.pth) you provided.
- Make sure your output format match the format we want.

```
file_name,pred1,pred2,pred3,pred4,pred5
001.jpg,air hockey,ampute football,archery,arm wrestling,axe throwing
002.jpg,air hockey,ampute football,archery,arm wrestling,axe throwing
003.jpg,air hockey,ampute football,archery,arm wrestling,axe throwing
004.jpg,air hockey,ampute football,archery,arm wrestling,axe throwing
005.jpg,air hockey,ampute football,archery,arm wrestling,axe throwing
006.jpg,air hockey,ampute football,archery,arm wrestling,axe throwing
007.jpg,air hockey,ampute football,archery,arm wrestling,axe throwing
008.jpg,air hockey,ampute football,archery,arm wrestling,axe throwing
009.jpg,air hockey,ampute football,archery,arm wrestling,axe throwing
010.jpg,air hockey,ampute football,archery,arm wrestling,axe throwing
```

Your pred_{student_id}.csv should looks like.

Submission

- Your submission should contain:

  - Network Structure: model.py

  - Train: train.py

  - Test: test.py

  - [other python files you add]

  - Model Weight: w_{student_id}.pth

  - Report : {student_id}.pdf

```
hw1_{student_id}.zip
├ model.py
├ train.py
├ test.py
├ [other python files you add]
├ w_{student_id}.pth
└ {student_id}.pdf
```

- **Compress all files into a single zip file named hw1_{student_id}.zip without any extra folder layers.**
- **Make sure not to include unnecessary files or folders (e.g., .DS_Store, __MACOSX). The submission must not contain the dataset or any .csv files.**

Penalty

- Format penalty - **20 points**

  - If you have any incorrect file format or name, then you will get **-20 points.**

- Late penalty - **10% per day**
- **Deadline: 2025/10/5 23:59**

Note

- If you have any questions, please contact us via the E3 system and include **ALL** TAs in your message